M101 · Unit 11 · Quick reference

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Recall **Grid**	$h = (b - a) / n, \quad x_i = a + ih$. Positive integer n panels; n+1 endpoints.
Recall **Endpoint sums**	$L_n = h \sum_{i=0}^{n-1} f(x_i)$; $R_n = h \sum_{i=1}^{n} f(x_i)$.
Recall **Midpoints**	$M_n = h \sum_{i=0}^{n-1} f(a + (i + \frac{1}{2})h)$.
Recall **Trapezoids**	$T_n = h[\frac{1}{2}f(a) + \sum_{i=1}^{n-1} f(x_i) + \frac{1}{2}f(b)]$. $T_n = (L_n + R_n) / 2$.
Understand **Monotone bracket**	Forward bounds. Increasing: $L_n \le I \le R_n$; decreasing: reversed. Width $h \mid f(b) - f(a)\mid$. T error at most half-width.
Consult **Unequal widths**	$T = \sum_{i=0}^{n-1} (x_{i+1} - x_i)(y_i + y_{i+1}) / 2$. Coordinates increasing; use actual widths and consistent units.

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Understand **Signs and exactness**	Midpoint/trapezoid exact for linear f. Reverse bounds: M,T change sign. $L_n(b,a) = -R_n(a,b)$.
Understand **Refinement**	If $Q_n - I \approx C / n^2$, finer error $\approx \mid Q_{2n} - Q_n \mid / 3$. Estimate, not a bound; inspect features and successive differences.
Recall **Error measures**	Absolute: $\lvert Q - I \rvert$. Relative: $\lvert Q - I \mid / \mid I \rvert$, only $I \neq 0$; use absolute tolerance near cancellation.
Consult **Before accepting**	Domain, units, hand sum, exact benchmark, refinement, sample positions; split known changes using one-sided piece formulas.
Understand **Limits of refinement**	Interpolated rows add no observations. Shared bias and missed features survive refinement. Retain precision until display.
Consult **Applicability**	Finite continuous or piecewise-continuous models on bounded intervals; inspect domains and split known changes.

Consult · Python

```
# Core formulas after validating n and finite inputs
h = (b - a)/n
mid = h*sum(f(a + (i + 0.5)*h) for i in range(n))
trap = h*((f(a) + f(b))/2
          + sum(f(a + i*h) for i in range(1, n)))
# Table with nonuniform increasing coordinates:
total = sum((x[i+1]-x[i])*(y[i]+y[i+1])/2
            for i in range(len(x)-1))
```

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